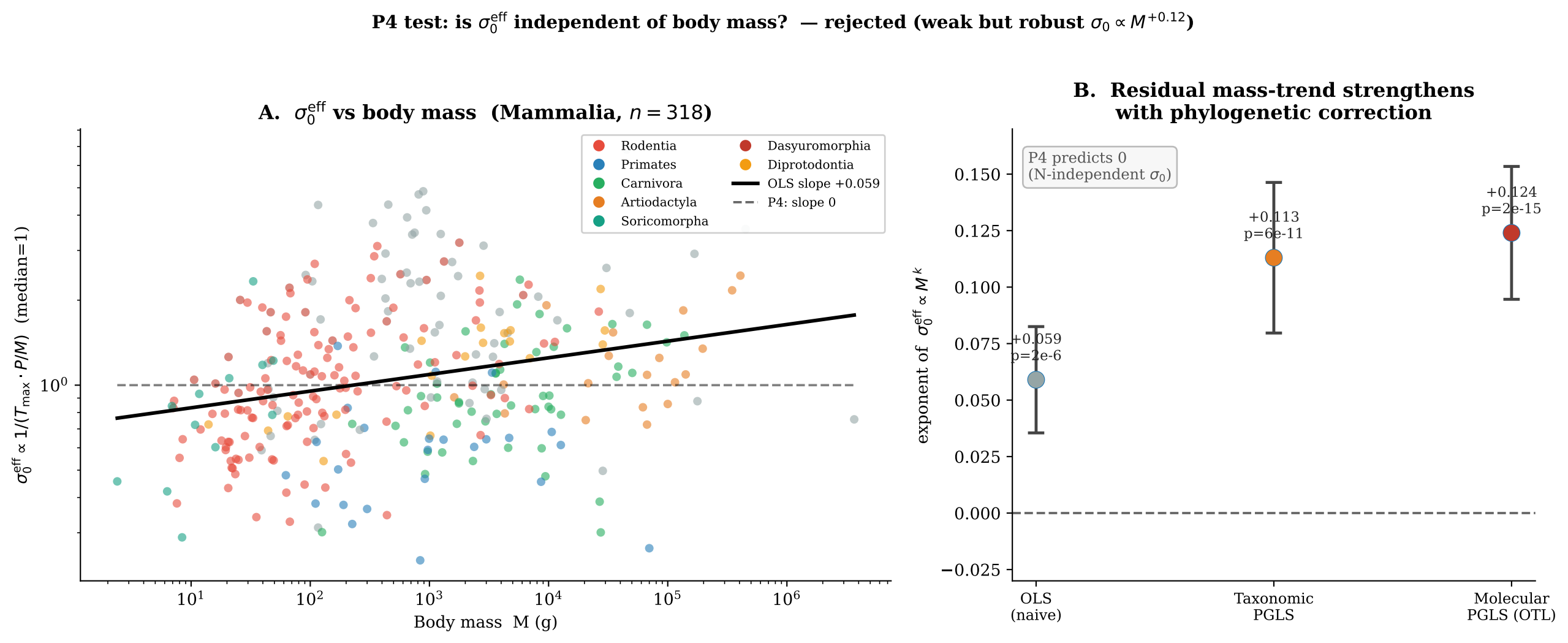

P4 test: is σ_0^{eff} independent of body mass? — rejected (weak but robust $\sigma_0 \propto M^{+0.12}$)
A. σ_0^{eff} vs body mass (Mammalia, $n = 318$)
Rodentia
Primates
Carnivora
Artiodactyla
Soricomorpha
Dasyuromorphia
Diprotodontia
OLS slope +0.059
P4: slope 0
$\sigma_0^{\text{eff}} \propto 1/(T_{\max} \cdot P/M)$ (median=1)
10^0
10^1
10^2
10^3
10^4
10^5
10^6
Body mass M (g)
B. Residual mass-trend strengthens with phylogenetic correction
P4 predicts 0
(N-independent σ_0)
exponent of $\sigma_0^{\text{eff}} \propto M^k$
0.150
0.125
0.100
0.075
0.050
0.025
0.000
−0.025
+0.059
p=2e-6
+0.113
p=6e-11
+0.124
p=2e-15
OLS
(naive)
Taxonomic
PGLS
Molecular
PGLS (OTL)